State-of-the-Art: Prediction of Resilient Modulus of Unsaturated Subgrade Soils

Zhong Han, S.M.ASCE[1]; and Sai K. Vanapalli, Ph.D., M.ASCE[2]

Abstract: In this paper, equations that were proposed in the literature over the past four decades to estimate or predict the variation of the resilient modulus with respect to soil suction for pavement base-course materials and subgrade soils are summarized into three groups: (1) empirical relationships, (2) constitutive models incorporating the soil suction into applied shearing or confining stresses, and (3) constitutive models extending the independent stress state variable approach. Two equations selected from each of the groups (a total of six equations) are used to predict the resilient modulus–soil suction correlations for three compacted subgrade soils. Strengths and limitations of these widely used equations are discussed based on the comparisons between the measurements and predictions. The key objective of the state-of-the-art research summarized in this paper is for assisting practicing engineers to choose suitable equations for the rational prediction of the resilient modulus taking into account the influence of the soil suction. **DOI: 10.1061/(ASCE)GM.1943-5622.0000631.** *© 2016 American Society of Civil Engineers.*

Author keywords: Pavement; Resilient modulus; Soil suction; Prediction.

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

Introduction

Pavements are the most widely used geotechnical structures in land transportation systems throughout the world. The structure of a typical pavement foundation is formed with compacted granular materials placed over compacted subgrade soils (Brown 1996). Two key criteria that are used in the design of pavement layers are the fatigue cracking at the bottom of the surface layer and the permanent deformation at the surface of the subgrade soils (Fig. 1). The fatigue cracking failure is found closely related to the resilient behavior of the pavement materials in response to the traffic loading (Seed et al. 1962).

Traffic loading in pavement analysis is conventionally simplified as a cyclic deviator stress (σ_d). The repeated application of the σ_d results in permanent strain (ε_p) and resilient strain (ε_r) within pavement materials, as shown in Fig. 2. The resilient modulus (M_R), which was initially recommended as per the AASHTO (1986) pavement design guide, has been widely used since then as the key soil property in the mechanistic pavement design methods to rationally characterize the resilient behavior of the pavement materials, analyze the fatigue failure of the surface layer, and dimension the multilayer system of the pavement structure. The definition of M_R has been introduced in the literature by Seed et al. (1962) as the ratio of the σ_d to the ε_r

$$M_R = \frac{\sigma_d}{\varepsilon_r} \tag{1}$$

Compacted pavement base/subbase materials and subgrade soils typically stay in an unsaturated condition and are subjected to environment influences. Environmental factors contribute to moisture regime and soil suction (ψ) variations within the pavement structure which, in turn, influence the strength and stiffness of pavement materials. Several research studies in recent years have demonstrated the strong correlations between the mechanical properties of unsaturated soils and the ψ (Fredlund et al. 1978; Edil et al. 1981; Wu et al. 1984; Alonso et al. 1990, 2013; Vanapalli et al. 1996; Drumm et al. 1997; Khalili et al. 2004; Ng and Yung 2008; Sawangsuriya et al. 2009a, b; Khosravi and McCartney 2012; Oh and Vanapalli 2013; Lu and Kaya 2014; Hoyos et al. 2015). It is therefore recommended to use ψ as a key parameter to interpret and predict the response of the mechanical behavior of unsaturated soils to the soil moisture regime fluctuations. Experimental investigations in recent years highlighted the influence of the ψ on the M_R of various pavement materials (Khoury and Zaman 2004; Yang et al. 2008; Thom et al. 2008; Khoury et al. 2011, 2012; Ng et al. 2013; Sivakumar et al. 2013; Ng and Zhou 2014; Abu-Farsakh et al. 2015; Salour and Erlingsson 2015). However, reliable determination of the M_R using experimental methods with soil suction control is time-consuming and needs sophisticated equipment as well as trained personnel. Because of these reasons, several researchers proposed various equations in the literature to predict the M_R–ψ correlations (Lytton 1995; Yang et al. 2005; Liang et al. 2008; Sawangsuriya et al. 2009a; Cary and Zapata 2011; Ng et al. 2013; Ba et al. 2013; Nokkaew et al. 2014; Han and Vanapalli 2015). These equations can be used in pavement design to take into account the influence of environmental factors.

The influence of several factors on the M_R of granular materials and subgrade soils are well documented in the literature (LeKarp et al. 2000; George 2004). However, a comprehensive state-of-the-art review of the equations for interpreting and predicting the variation of the M_R with respect to the ψ for pavement materials remains outstanding. In this paper, various equations available in the literature that were proposed in this direction are summarized into three groups: (1) empirical relationships, (2) constitutive models incorporating the soil suction into applied shearing or confining stresses, and (3) constitutive models extending the independent stress state variable approach. Two equations from each of the groups are selected to predict the variation of the M_R with respect to ψ for three compacted subgrade soils. The

[1]Ph.D. Candidate, Dept. of Civil Engineering, Univ. of Ottawa, Ottawa, Canada. E-mail: zhan010@uottawa.ca

[2]Professor, Dept. of Civil Engineering, Univ. of Ottawa, Ottawa, Canada (corresponding author). E-mail: vanapall@eng.uottawa.ca

Note. This manuscript was submitted on April 18, 2014; approved on October 30, 2015; published online on January 11, 2016. Discussion period open until June 11, 2016; separate discussions must be submitted for individual papers. This paper is part of the ***International Journal of Geomechanics***, © ASCE, ISSN 1532-3641.

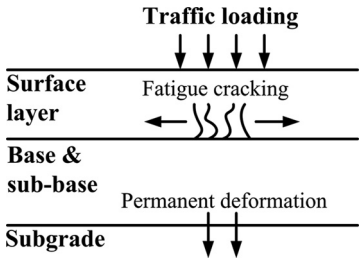


Fig. 1. Schematic of a typical asphalt pavement structure

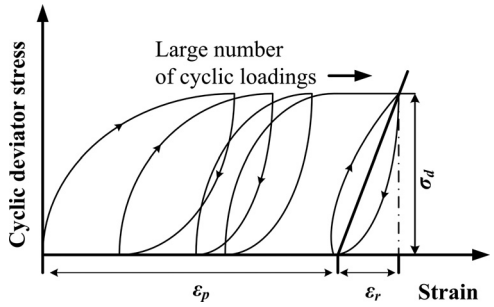


Fig. 2. Response of soils to cyclic deviator stress

predictions are compared with the measured results. In addition, predictions for one subgrade soil by the six selected equations are plotted as surfaces presenting resilient modulus–cyclic stress–soil suction relationships for more rigorous comparisons. Strengths and limitations of the three summarized equation groups are reviewed based on the comparison studies. The state-of-the-art summary and comparison studies presented in this paper are helpful for assisting practicing engineers to choose appropriate equations for the reliable prediction of the resilient modulus taking into account the influence of the soil suction.

Measurement of Resilient Modulus–Soil Suction Correlations

Repeated load triaxial tests (RLTTs) are widely used to determine the M_R of pavement materials (Uzan 1985; Tian et al. 1998). There are a variety of standard testing protocols that can be followed to determine the M_R [e.g., AASHTO T292-91 (AASHTO 1992), AASHTO T307-99 (AASHTO 2003), and NCHRP 1-28A (Witczak 2003)]. These testing protocols have certain differences with respect to sequences of applying loading stresses, measurement of deformations, specimen preparation, and conditioning methodology before and during the testing procedure. More detailed information of the various testing protocols are summarized by Puppala (2008). Several researchers incorporated various soil suction measurement or control techniques into the RLTTs to investigate the influence of the ψ on the M_R of pavement materials. For example, the ψ of the specimens can be measured after the completion of RLTTs using the filter paper method or null pressure-plate method, which are both simple and inexpensive (Fredlund et al. 1977; Yang et al. 2005; Liang et al. 2008; Azam et al. 2013). However, such suction measurement methods are user dependent and sensitive to testing protocols (Power et al. 2008; Power and Vanapalli 2010). Also, the applied stress on the specimens influences the soil suction (Vanapalli et al. 1999; Ng and Pang 2000). Because of the differences in the applied stress levels between the RLTTs and the soil suction measurement tests, the measured soil suction may not precisely represent the soil suction within the specimens during the RLTTs.

In recent years, modified cyclical triaxial testing systems equipped with various techniques (thermal dissipation sensor, psychrometer, suction probe, axis-translation technique, etc.) are used to measure or control the soil suction during the RLTTs to obtain reliable test results on the variation of the M_R with soil suction (Gupta et al. 2007; Yang et al. 2008; Craciun and Lo 2010; Cary and Zapata 2011; Ng et al. 2013; Sivakumar et al. 2013). Such modified triaxial testing systems are expensive, cumbersome, and require trained professionals to conduct the tests. In addition, several weeks are typically required to achieve equilibrium conditions with respect to each applied suction value for fine-grained unsaturated soils if the axis-translation technique is employed. (Khoury et al. 2011; Ng et al. 2013).

Prediction of Resilient Modulus–Soil Suction Correlations

The Mechanistic-Empirical Pavement Design Guide (MEPDG 2004) (ARA, Inc., ERES Consultants Division 2004) recommends using Eq. (2) to estimate the M_R at optimum moisture content (OMC) condition (M_{ROPT}) taking into account the influence of the applied stresses

$$M_R = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(\frac{\tau_{oct}}{p_a} + 1\right)^{k_3} \tag{2}$$

where p_a = atmosphere pressure (i.e., 101.3 kPa); θ_b = bulk stress; τ_{oct} = octahedral shear stress; and k_1, k_2, and k_3 = model parameters.

Eq. (3) is used in MEPDG to calibrate the M_{ROPT} values considering the influence of seasonal moisture content fluctuations

$$\log\left(\frac{M_R}{M_{ROPT}}\right) = a + \frac{b-a}{1+\exp\left[\ln\left(-\frac{b}{a}\right) + k_m(S - S_{OPT})\right]} \tag{3}$$

where S = degree of saturation (in decimals); S_{OPT} = degree of saturation at OMC condition (in decimals); a = minimum of log (M_R/M_{ROPT}); b = maximum of log (M_R/M_{ROPT}); and k_m = regression parameter. Parameter values $a = -0.5934$, $b = 0.4$, and $k_m = 6.1324$ are suggested for fine-grained soils, and parameter values $a = -0.3123$, $b = 0.3$, and $k_m = 6.8157$ are suggested for coarse-grained soils.

Eq. (3) is an empirical relationship that was derived from extensive experimental data (Zapata et al. 2007). This equation is widely implemented in practice because it (1) uses parameters a, b, and k_m with specific values that do not need calibration and (2) only requires the information of the M_{ROPT} and S that are conventionally determined. The variations of the (M_R/M_{ROPT}) with respect to the (S–S_{OPT}) predicted by Eq. (3) for fine-grained and coarse-grained soils are shown in Fig. 3.

Recent studies show that the M_R–moisture content relationship is highly soil-type–dependent; hence, it is difficult to be precisely characterized using either empirical or mechanistic methods (Heath et al. 2004; Ng et al. 2013). Sauer and Monismith (1968), Edris and Lytton (1976), Fredlund et al. (1977), and Edil and Motan (1979) were the pioneers who suggested using soil suction as the fundamental parameter to interpret the response of resilient behavior of pavement materials to soil moisture content variations. Over the last four decades, various equations have been proposed in the literature

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

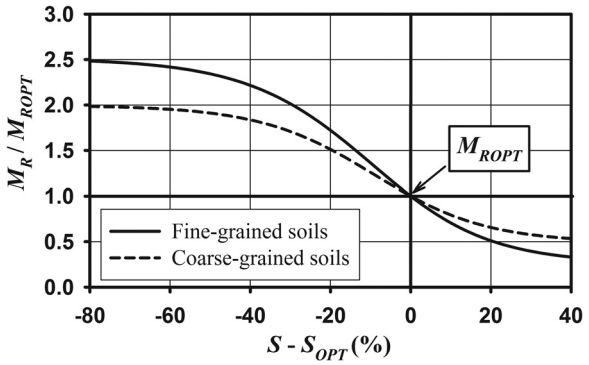


Fig. 3. Predicted M_R/M_{ROPT} with respect to variation in moisture content by Eq. (3)

to predict the M_R–ψ correlations for different pavement materials. These equations can be categorized into three groups (Groups A, B, and C) based on the different approaches used to account for the influence of the soil suction.

Group A: Empirical Relationships

Empirical relationships are simple equations used to simulate the measured M_R–ψ correlations for specific soils. Such equations do not provide theoretical justification but are developed based on statistical analysis performed on a large database of experimental results for soils with similar properties. Some empirical equations available in the literature are summarized below [soils are classified as per the AASHTO M-145 (AASHTO 2008) soil classification system].

Johnson et al. (1986) proposed Eq. (4), which was derived for sandy soils

$$M_R = 1.35 \times 10^6 \times (101.36 - \psi)^{2.36} (J_1)^{3.25} (\gamma_d)^{3.06} \tag{4}$$

where J_1 = first stress invariant; γ_d = dry density; and M_R is in MPa, ψ and J_1 are in kPa, and γ_d is in Mg/m^3.

Parreira and Gonçalves (2000) proposed Eq. (5), which was derived for a lateritic soil (A-7-6) from Brazil over the soil suction range of 0–87,500 kPa

$$M_R = 14.10 \sigma_d^{0.782} \psi^{0.076} \tag{5}$$

where M_R is in MPa, σ_d is in kPa, and ψ is in kPa.

Ceratti et al. (2004) proposed Eq. (6), which was derived for a lateritic soil (A-7-6) from Brazil over an in situ soil suction range of 0–14 kPa

$$M_R = 142 + 16.9\psi \tag{6}$$

where M_R is in MPa and ψ is in kPa.

Doucet and Doré (2004) proposed Eq. (7), which was derived for several partial crushed and crushed granular materials from Quebec, Canada

$$M_R = 1060\theta_b - 8700\psi + 57000 \tag{7}$$

where ψ and M_R are in kPa.

Sawangsuriya et al. (2009a) proposed Eqs. (8) and (9), which were derived for four fine-grained subgrade soils (two A-4 soils and two A-7-6 soils) from Minnesota over the soil suction range of 0–10,000 kPa

$$M_R/M_{RSAT} = -5.61 + 4.54 \log(\psi) \tag{8}$$

$$M_R/M_{ROPT} = -0.24 + 0.25 \log(\psi) \tag{9}$$

where $M_{RSAT} = M_R$ at saturation condition; and ψ is in kPa.

Ba et al. (2013) proposed Eq. (10), which was derived for four unbound granular base materials from Senegal over the soil suction range of 0–100 kPa

$$M_R/M_{ROPT} = 0.385 + 0.267 \log(\psi) \tag{10}$$

where ψ is in kPa.

Note that Eq. (3) was also developed following this approach to relate M_R with S. When the soil-water characteristic curve (SWCC), which describes the S–ψ relationships, is known, Eq. (3) can also be extended to predict M_R–ψ correlations. Empirical equations are widely used in practice, because they are easy to apply. For example, for prediction, Eqs. (8)–(10) only require the experimental data of M_{ROPT} or M_{RSAT}, which can be determined using conventional equipment without soil suction control, and no additional tests are required.

Several investigators proposed constitutive models to predict the M_R by relating applied stresses using model parameters. The power law equations proposed by Moossazadeh and Witczak (1981) [Eq. (11)], Uzan (1985) [Eq. (12)], and MEPDG 2004 [Eq. (2)] are the commonly used constitutive models

$$M_R = k_1 \left(\frac{\sigma_d}{p_a}\right)^{k_2} \tag{11}$$

$$M_R = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(\frac{\tau_{oct}}{p_a}\right)^{k_3} \tag{12}$$

Eqs. (2), (11), and (12) and other equations developed following similar formulations can be revised to take into account the influence of soil suction using two different approaches. One approach is to include the contribution of the soil suction into applied confining or shearing stresses (Group B); the other approach is based on using the independent stress state variable approach (Group C).

Group B: Constitutive Models Incorporating the Soil Suction into Applied Shearing or Confining Stresses

Several constitutive models available in the literature that incorporate the soil suction into applied shearing or confining stresses are summarized below.

Loach (1987) proposed Eq. (13), which was derived for three fine-grained soils from the United Kingdom over the soil suction range of 0–100 kPa

$$M_R = \frac{\sigma_d}{k_1} \left[\frac{c\sigma_c + \psi}{\sigma_d}\right]^{k_2} \tag{13}$$

Jin et al. (1994) proposed Eq. (14), which was derived for two granular base materials from Rhode Island

$$\Delta M_R = K_1 K_2 \theta_b^{K_2 - 1} (\Delta\theta_{bT} + \Delta\theta_{b\psi}) \tag{14}$$

Lytton (1995) proposed Eq. (15) for granular base materials. Eq. (15) was further verified using experimental data derived from nine granular base materials from Texas by Gu et al. (2014)

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

$$M_R = k_1 p_a \left(\frac{\theta_b - 3f\theta\psi}{p_a}\right)^{k_2} \left(\frac{\tau_{oct}}{p_a}\right)^{k_3} \quad (15)$$

Heath et al. (2004) proposd Eq. (16), which was derived for a typical granular base material from California

$$M_R = k_1 p_a \left[\frac{(\theta_b/3) - u_a + \chi\psi}{p_a}\right]^{k_2} \left(\frac{\sigma_d}{p_a}\right)^{k_3} \quad (16)$$

Yang et al. (2005) proposd Eq. (17), which was derived for two fine-grained subgrade soils (one A-7-5 soil and one A-7-6 soil) from Taiwan, China, over the soil suction range of 0–10,000 kPa

$$M_R = k_1(\sigma_d + \chi\psi)^{k_2} \quad (17)$$

Liang et al. (2008) proposed Eq. (18), which was derived for two fine-grained subgrade soils (one A-4 soil and one A-6 soil) over the suction range of 150–380 kPa. Eq. (18) was also validated using eight sets of experimental data on fine-grained soils from the literature

$$M_R = k_1 p_a \left(\frac{\theta_b + \chi\psi}{p_a}\right)^{k_2} \left(1 + \frac{\tau_{oct}}{p_a}\right)^{k_3} \quad (18)$$

Oh et al. (2012) proposed Eq. (19), which was derived for granular base and subgrade soils from Florida

$$M_R = k_1 p_a \left(\frac{\theta_b + 3k_4\psi\theta}{p_a}\right)^{k_2} \left(\frac{\tau_{oct}}{p_a} + 1\right)^{k_3} \quad (19)$$

Sahin et al. (2013) proposed Eq. (20) for granular base materials

$$M_R = k_1 p_a \left[\frac{\theta_b - 3f\theta(\psi_0 + \beta\frac{\theta_b}{3} + \alpha\tau_{oct})}{p_a}\right]^{k_2} \left(\frac{\tau_{oct}}{p_a} + 1\right)^{k_3} \quad (20)$$

where K_1, K_2, k_1, k_2, k_3, and k_4 = model parameters; c = compressibility factor; σ_c = confining stress; ΔM_R = changes in M_R; $\Delta\theta_{b\psi}$ = changes in θ_b due to soil suction; $\Delta\theta_{bT}$ = changes in θ_b due to temperature; θ = volumetric water content; f = saturation factor ($1 < f < 1/\theta$); u_a = pore-air pressure; χ = Bishop's effective stress parameter; ψ_0 = initial soil suction; and α and β = Henkel pore-water pressure parameters.

These summarized models were developed based on either Bishop's effective stress concept or unsaturated soils [Bishop 1959; for example, Eqs. (16)–(18)] or micromechanics theory and thermodynamic laws [Fung 1977; for example, Eqs. (14), (15), (19), and (20)].

Group C: Constitutive Models Extending the Independent Stress State Variable Approach

Several constitutive models available in the literature, which consider the soil suction as a stress state variable that influences the mechanical behavior of unsaturated soils independently, are summarized below.

Fredlund et al. (1977) proposed Eq. (21), which was developed for a glacial till from Saskatchewan, Canada, over the soil suction range of 0–1,000 kPa. In Eq. (21), c_{ld} = intercept; m_{ld} = slope of the σ_d versus log (M_R) relationship; and c_{ld} and m_{ld} = functions of ψ

$$\log M_R = c_{ld} - m_{ld}(\sigma_d) \quad (21)$$

Oloo and Fredlund (1998) proposed Eq. (22) for coarse-grained soils and Eqs. (23) and (24) for fine-grained soils

$$M_R = k\theta_b^{m_b} + k_s\psi \quad (22)$$

$$M_R = k_2 - k_3(k_1 - \theta_b) + k_s\psi \text{ when } k_1 > \theta_b \quad (23)$$

$$M_R = k_2 + k_4(\theta_b - k_1) + k_s\psi \text{ when } k_1 < \theta_b \quad (24)$$

Gupta et al. (2007) proposed Eqs. (25) and (26), which were derived for four fine-grained subgrade soils (two A-4 soils and two A-7-6 soils) from Minnesota, over the soil suction range of 10–10,000 kPa

$$M_R = k_1 p_a \left(\frac{\theta_b - 3k_4}{p_a}\right)^{k_2} \left(k_5 + \frac{\tau_{oct}}{p_a}\right)^{k_3} + \alpha_1\psi^{\beta_1} \quad (25)$$

$$M_R = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(1 + \frac{\tau_{oct}}{p_a}\right)^{k_3} + k_{us} p_a \Theta^{\kappa}\psi \quad (26)$$

Khoury et al. (2009) proposed Eq. (27), which was derived for several subgrade soils (ranging from A-4 to A-7) from Oklahoma over the soil suction range of 0–6,000 kPa

$$M_R = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(k_4 + \frac{\tau_{oct}}{p_a}\right)^{k_3} + \alpha_1\psi^{\beta_1} \quad (27)$$

Caicedo et al. (2009) proposed Eq. (28), which was derived for three nonstandard granular base materials from Andes Cordillera, Colombia, over the soil suction range of 0–200 kPa

$$M_R = k_1 p_a \left(1 + k_2\frac{\sigma_d}{p_a}\right)\left(\frac{\psi}{p_a}\right)^{k_3} \frac{f(e)}{f(0.33)} \quad (28)$$

Khoury et al. (2011) proposed Eq. (29), which was derived for a manufactured soil with soil suction levels similar to a silty soil. The hysteresis behavior in the M_R with respect to soil suction was considered within the soil suction range of 0–100 kPa

$$M_R = \left[k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(1 + \frac{\tau_{oct}}{p_a}\right)^{k_3} + (\psi - \psi_0) \times \left(\frac{\theta_d}{\theta_s}\right)^{\left(\frac{1}{n}\right)}\right] \times \left(\frac{\theta_d}{\theta_w}\right) \quad (29)$$

Cary and Zapata (2011) proposed Eq. (30), which was derived for a granular soil (A-1-a) and a clayey sand (A-4) from Arizona over the soil suction range of 0–250 kPa. Eq. (30) was further verified using experimental data of two sandy subgrade soils (one A-4 soil and one A-2-4 soil) from Sweden over the soil suction range of 0–450 kPa by Salour et al. (2014)

$$M_R = k_1 P_a \left(\frac{\theta_{net} - 3\Delta u_{w\text{-}sat}}{p_a}\right)^{k_2} \left(\frac{\tau_{oct}}{p_a} + 1\right)^{k_3} \left(\frac{\psi_0 - \Delta\psi}{p_a} + 1\right)^{k_4} \quad (30)$$

Ng et al. (2013) proposed Eq. (31), which was derived for a subgrade soil (A-7-6) from Hong Kong, China, over the soil suction range of 0–250 kPa

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

$$M_R = M_0\left(\frac{p}{p_r}\right)^{k_1}\left(1+\frac{q_{\text{cyc}}}{p_r}\right)^{k_2}\left(1+\frac{\psi}{p}\right)^{k_3} \tag{31}$$

Azam et al. (2013) proposed Eq. (32), which was derived for four recycled, unbound granular materials from Australia over the soil suction range of 0–10 kPa

$$M_R = k\left(\frac{\sigma_m}{p_a}\right)^{k_1}\left(\frac{\tau_{\text{oct}}}{\tau_{\text{ref}}}\right)^{k_2}\left(\frac{\psi}{p_a}\right)^{k_3}\left[\frac{\text{DDR}(1-k_4\text{RCM}/100)}{100}\right]^{k_5} \tag{32}$$

Han and Vanapalli (2015) proposed Eq. (33), which was derived based on the experimental results of 11 different compacted, fine-grained subgrade soils that were tested following various protocols. Model parameter ξ was found equal to 2.0 for all the examined subgrade soils

$$\frac{M_R - M_{R\text{SAT}}}{M_{R\text{OPT}} - M_{R\text{SAT}}} = \frac{\psi}{\psi_{\text{OPT}}}\left(\frac{S}{S_{\text{OPT}}}\right)^{\xi} \tag{33}$$

where k, k_s, k_{us}, k_1, k_2, k_3, k_4, k_5, α_1, β_1, κ, m_b, and ξ = model parameters; $\Theta = \theta/\theta_s$ = normalized volumetric water content; θ_s = saturated volumetric water content; e = void ration, $f(e) = (1.93-e)^2/(1+e)$; θ_d = volumetric water content along the drying curve; θ_w = volumetric water content along the wetting curve; n = model parameter obtained from the Fredlund and Xing (1994) SWCC model; $\theta_{\text{net}} = \theta_b - 3u_a$ = net bulk stress; $\Delta u_{w\text{-sat}}$ = buildup of pore-water pressure under saturated conditions; $\Delta\psi$ = relative change of soil suction with respect to ψ_0 due to buildup of pore-water pressure under unsaturated conditions; $p = (\theta_b/3 - u_a)$ = net mean stress; p_r = reference pressure = 1 kPa; q_{cyc} = cyclic shear stress; $M_0 = M_R$ value at reference stress state when $(u_a - u_w) = 0$, $(p - u_a) = p_r$, and $q_{\text{cyc}} = p_r$; DDR = dry density ratio (%); RCM = percent of recycled clay masonry (%); τ_{ref} = reference shear stress; σ_m = mean normal stress = $\theta_b/3$; and ψ_{OPT} = soil suction at OMC.

Model parameters with the same notations (e.g., k_1, k_2, k_3) are used in Eqs. (2), (11), (12), and (13)–(33) for simplicity purposes. These model parameters in different equations, however, may have different physical meanings as well as values.

It should be noted that apart from the M_R, the volumetric strain (ε_v) and the deviator strain (ε_q) under cyclic loading can also be used to characterize the resilient behavior of pavements. The variations of ε_v and ε_q with ψ can also be predicted using the approaches described in Groups B and C. Prediction models for ε_v and ε_q, however, are beyond the scope of this paper. More discussions on this topic are available in Nowamooz et al. (2011) and Ho et al. (2014a, b).

Some common concerns with respect to the prediction equations are discussed below:

1. The M_R–ψ correlation of a specific pavement material tested following certain techniques can be reasonably simulated using a simple empirical equation. However, there are uncertainties using the same empirical equation for other soils and other testing scenarios. Also, empirical equations are only suitable for a certain soil suction range; prediction results beyond this range may not be reliable. For example, at saturation condition when $\psi = 0$, Eq. (5) predicts $M_{R\text{SAT}} = 0$, whereas Eqs. (8)–(10) predict $M_{R\text{SAT}}$ values equal to negative infinity. Because of such limitations, empirical equations are not suggested for rigorous analysis or numerical simulations as they likely contribute to instability problems. The constitutive models are comparatively more flexible to reproduce different M_R–ψ correlations using various model parameters. Reliable calibration of the model parameters for different pavement materials is based on performing regression analysis on large amounts of experimental data, which could be time-consuming and expensive.
2. Most of the summarized constitutive models are revisions of Eqs. (2), (11), or (12). At saturation condition ($\psi = 0$), some of these constitutive models reduce to Eqs. (2), (11), and (12), or similar formulations, and predict $M_{R\text{SAT}}$. For instance, when $\psi = 0$, Eq. (17) is similar to Eq. (11); Eqs. (15) and (16) are similar to Eq. (12); and Eqs. (18)–(20), (26), (29), and (30) are similar to Eq. (2). These revised equations can be represented using the following formulations:

$$M_R = \begin{cases} M_{R\text{SAT}} \times f_1(\psi), & f_1(0) = 1 \\ M_{R\text{SAT}} + f_2(\psi), & f_2(0) = 0 \end{cases} \tag{34}$$

where $f_1(\psi)$ and $f_2(\psi)$ = functions representing the contribution of the soil suction toward the $M_{R\text{SAT}}$. In the pavement construction practice, materials are normally compacted at or near OMC to achieve maximum density and assure better performance (Zaman et al. 2010). Eqs. (2), (11), and (12) were proposed and are generally used for $M_{R\text{OPT}}$. Several studies, based on various pavement materials, suggested specific values and estimation approaches for model parameters k_1, k_2, and/or k_3 in Eqs. (2), (11), and (12) to predict the $M_{R\text{OPT}}$ (e.g., Hossain 2008; Stolle et al. 2009; Caliendo 2012; and Shaqlaih et al. 2013). However, the model parameters for the $M_{R\text{OPT}}$ are not suitable for the $M_{R\text{SAT}}$. Referring the $M_{R\text{OPT}}$ values for predicting M_R–ψ correlations is common in empirical equations [Eqs. (3), (8), and (10)], because $M_{R\text{OPT}}$ is conventionally determined. The same approach can also be extended for constitutive models. Eq. (33) allows the direct use of $M_{R\text{OPT}}$ for prediction. $M_{R\text{OPT}}$ can be expressed using any stress-dependent constitutive models [e.g., Eqs. (2), (10), and (11)] to take into account the influence of applied stresses on the M_R–ψ correlations (Han and Vanapalli 2015).
3. Incorporating the soil suction into applied stresses (Group B) is typically based on Bishop's effective stress concept for unsaturated soils using parameter χ or micromechanics theory and thermodynamic laws using parameter f. Determi-nation of the χ and f needs experimental investigations that could be cumbersome and expensive. Some investigators, therefore, suggested using empirical relationships to estimate χ and f (Khalili and Khabbaz 1998; Heath et al. 2004; Khalili et al. 2004; Yang et al. 2006; Liang et al. 2008; Nokkaew et al. 2014). Concerns are, however, raised, because using empirical relationships may introduce additional uncertainties into the predictions (Cary and Zapata 2011).

Model parameters used in summarized constitutive equations are different from those used in Eqs. (2), (11), and (12) both in terms of physical meaning and values. For example, the k_2 used in Eq. (11) describes the response of the M_R to the applied shearing stress σ_d; the k_2 used in Eq. (17), however, describes the response of M_R to a stress state composed of σ_d and ψ. The k_2 used in Eq. (2) describes the relationship between M_R and the applied confining stress θ_b; the k_2 used in Eq. (17), however, describes the relationship between M_R and a stress state composed of θ_b and ψ. The differences in the physical meaning of the model parameters not only translate to different values but also cause changes to other model parameters. Therefore, estimation methods of the model parameters used in Eqs. (2), (11), and (12), which were well established and

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

documented in the literature, may not be suitable for the equations categorized in Groups B and C.

4. The M_R–ψ correlations show a nonlinear increasing trend within lower suction range. Such a trend, however, may not be continued at higher suction range. Edil and Motan (1979) reported that the M_R of several clay and sand-clay mixtures from Wisconsin increased with soil suction up to approximately 800 kPa, beyond which the M_R started to decrease. Ceratti et al. (2004) reported that the M_R of a lateritic soil from Brazil increased with the soil suction up to 1,000 kPa. Further soil suction increases did not contribute to a significant increase in the M_R. Such behaviors of the M_R may be attributed to the more effective contribution of soil suction on wetted contacted area of the soil particles within the low soil suction range (boundary effect zone and the transition zone) to the soil strength and stiffness properties (Vanapalli et al. 1996). Within high soil suction range (residual zone), the moisture content is significantly low, and the wetted contact area of soil particles is limited. There is breakage of the capillary menisci, which limits or ceases the contribution of soil suction toward the M_R (Caicedo et al. 2009). In addition, hysteresis effects, which are also generally attributed to the influence of the moisture regime, are widely reported for the M_R of different pavement materials (Parreira and Gonçalves 2000; Khoury and Zaman 2004; Khoury et al. 2011; Ng et al. 2013). There are significant differences between the M_R values measured from specimens tested following the drying process and those measured following the wetting process. It is reasonable and necessary to incorporate the influence of the moisture regime in the prediction of the M_R–ψ correlations. The influence of the moisture regime is considered in some summarized equations by using parameters χ, f, θ, S, or Θ. Equations neglecting the influence of the moisture regime may encounter limitations, such as predicting a continuous increasing trend in M_R over the entire soil suction range and being incapable of taking into account the hysteresis effects.

Comparisons and Discussions

Two equations from each of the three groups—Eqs. (3) and (8) from Group A, which are empirical relationships; Eqs. (17) and (18) from Group B, which are constitutive models incorporating the soil suction into applied shearing or confining stresses; and Eqs. (27) and (31) from Group C, which are constitutive models extending the independent stress state variable approach—are chosen for comparison studies. Eq. (3) was developed from a large number of testing data on different soils and is one of the most widely used equations in pavement design practice; Eq. (8) well predicted the M_R of four representative subgrade soils from Minnesota (Sawangsuriya et al. 2009a); and Eqs. (17), (18), (27), and (31), on the other hand, are constitutive models verified with experimental data on a variety of soil types.

Experimental data derived from three different subgrade soils with similar soil properties [Yang et al. 2005; Gupta et al. 2007; and Ng et al. 2013; all the soils are classified as A-7-6 soils according to AASHTO M-145 (AASHTO 2008)] are selected to provide comparisons between the measured M_R and the predicted M_R using the selected equations. The soil properties, compaction characteristics, and experimental details of the three subgrade soils are summarized in Table 1. Fig. 4 shows the measured SWCC for the three soils (symbols) and the predicted SWCC (lines) using the Fredlund and Xing (1994) equation

$$S = 1/\{\ln[2.718 + (\psi/a)^n]\}^m \tag{35}$$

where a, m, and n = model parameters. The comparisons between the predicted and measured M_R values are performed following the combinations shown in Table 2.

Comparisons for Empirical Equations from Group A

Figs. 5–7 show the comparisons between the measured M_R and the predictions using empirical Eq. (3) and/or Eq. (8) for the pulverized

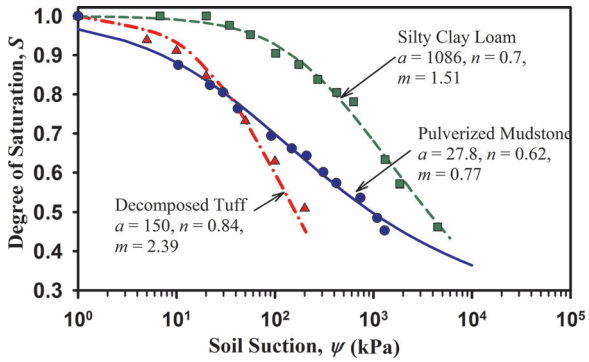


Fig. 4. SWCC of the three soils used in this study

Table 1. Details of the Three Soils Used in this Study

Detail	Pulverized mudstone from Taiwan	Decomposed tuff from Hong Kong	Silty clay loam from Minnesota
Source	Yang et al. (2005)	Ng et al. (2013)	Gupta et al. (2007)
Classification[a]	A-7-6	A-7-6	A-7-6
w_L (%)	50	43	42
I_p (%)	23	14	24
MDD (Mg/m^3)	18	17.6	15.8
OMC (%)	17	16.3	22
G_s	2.67	2.73	2.69
RC (%)	100	100	98
M_R testing procedure	AASHTO T292-91 (AASHTO 1992)	AASHTO T307-99 (AASHTO 2003)	NCHRP 1-28A (Witczak 2003)
Suction measurement	Filter paper method	Axis-translation technique	Thermal dissipation sensor
Shearing and confining stresses (kPa)	σ_d = 21, 48, 103; σ_c = 21	q_{cyc} = 30, 40, 55, 70; σ_{cnet} = 30	σ_d = 28, 49, 92; σ_c = 14

Note: w_L = liquid limit; I_p = plasticity index; MDD = maximum dry density; G_s = specific gravity; RC = relative compaction; σ_{cnet} = net confining stress = $(\sigma_c - u_a)$.

[a]Classified per AASHTO M-145 soil classification system (AASHTO 2008).

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

Table 2. Combinations of the Comparisons

Equation	Pulverized mudstone [Yang et al. (2005)]	Decomposed tuff [Ng et al. (2013)]	Silty clay loam [Gupta et al. (2007)]
Group A	Eqs. (3) and (8)	Eqs. (3) and (8)	Eq. (3)
Group B	Eq. (18)	Eqs. (17) and (18)	Eqs. (17) and (18)
Group C	Eqs. (27) and (31)	Eq. (27)	Eq. (31)

Note: Selected equations are not used to predict the measurements from the same study.

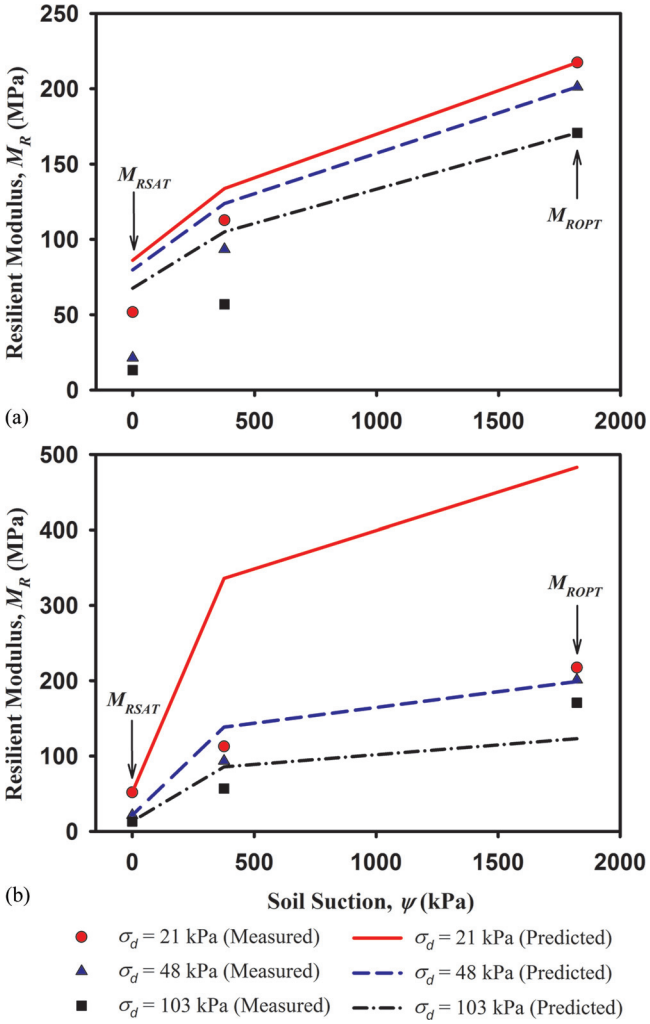


Fig. 5. Comparisons between the variation of the M_R with respect to soil suction of the PM measured by Yang et al. (2005) and predicted by: (a) Eq. (3), $R^2 = 0.9$; (b) Eq. (8), $R^2 = 0.6$

mudstone (PM), decomposed tuff (DT), and silty clay loam (SCL), respectively. Fig. 8 shows the overall predictions provided by the two equations.

The measured M_R–ψ correlations for the three soils show nonlinear characteristics and are reasonably predicted using Eqs. (3) and (8). Witczak et al. (2002) suggested subjective criteria for goodness predictions based on the coefficient of determination (R^2) values. An excellent fit can be defined as $R^2 \geq 0.9$, a good fit covers the R^2 range of 0.7–0.89, and a fair fit is defined as $0.4 \leq R^2 \leq 0.69$. According to this criteria, Eq. (3) provides excellent prediction for the PM and DT [Figs. 5(a) and 6(a)], good prediction for the SCL

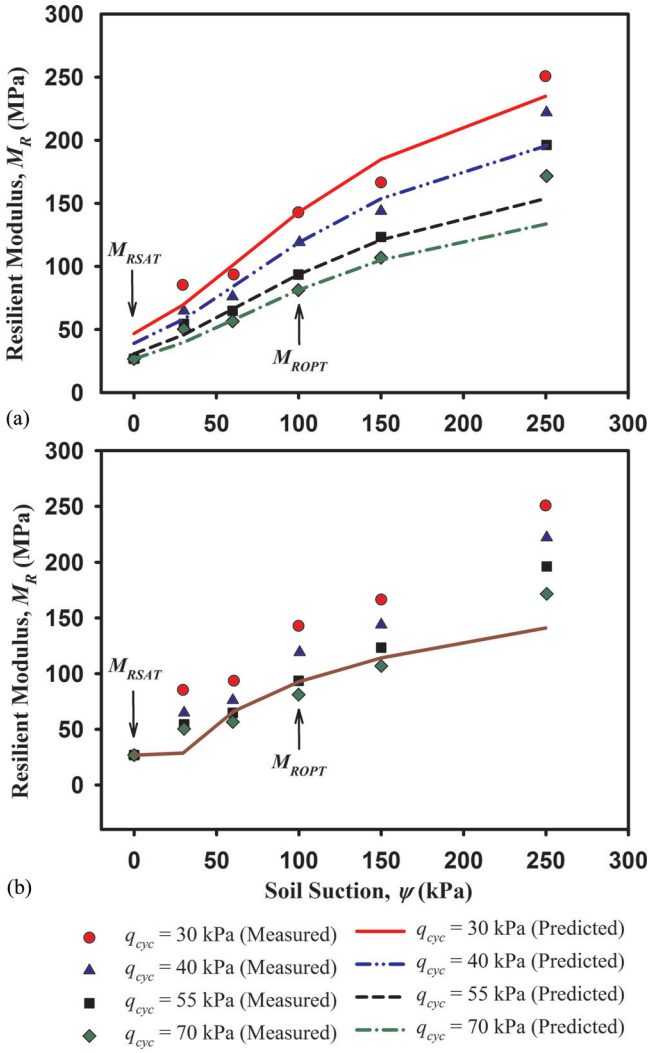


Fig. 6. Comparisons between the variations of the M_R with respect to soil suction of the DT measured by Ng et al. (2013) and predicted by: (a) Eq. (3), $R^2 = 0.95$; (b) Eq. (8), $R^2 = 0.82$

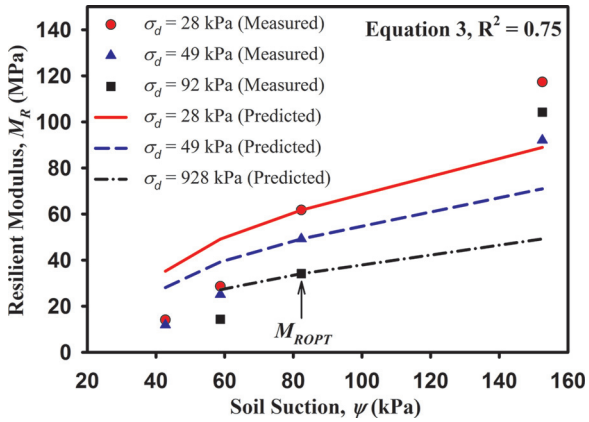


Fig. 7. Comparisons between the variations of the M_R with respect to soil suction of the SCL measured by Gupta et al. (2007) and predicted by Eq. (3)

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

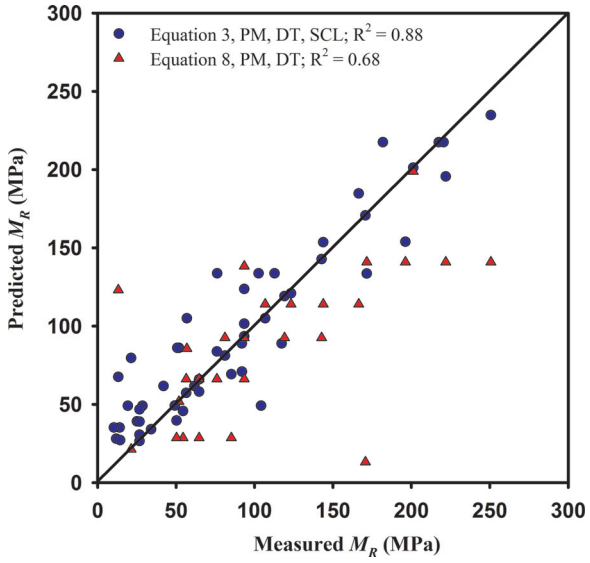


Fig. 8. Overall predictions provided by Eqs. (3) and (8) in comparison with the measured values

(Fig. 7), and an overall good prediction (Fig. 8). It is observed that Eq. (3) tends to overpredict the M_R values of the PM and SCL on the wet side of the OMC (in which soil suction is lower than the OMC condition) and underpredict the M_R values of the SCL for the dry side of the OMC (in which soil suction is higher than the OMC condition). Eq. (3) tends to overpredict most of the experimental data, as shown in Fig. 8.

Eq. (8) provides fair prediction for the PM [Fig. 5(b)], good prediction for the DT [Fig. 6(b)], and an overall fair prediction (Fig. 8). It is observed that the predictions of Eq. (8) are sensitive to the M_{RSAT}. Small differences in the M_{RSAT} contribute to significant variations in the predicted M_R at higher soil suction values, as shown in Fig. 5(b). On the other hand, the measured M_{RSAT} of the DT is a constant irrespective of the different cyclic stresses (q_{cyc}). The measured M_R at other soil suction levels is influenced by the q_{cyc} (Ng et al. 2013). The predictions do not reflect the influence of the applied stresses but are identical at different q_{cyc} values [Fig. 6(b)]. Eq. (8) tends to underpredict most of the experimental data, as shown in Fig. 8.

Comparisons for Constitutive Models from Group B

Yang et al. (2006) and Liang et al. (2008) suggested estimating the χ values in Eqs. (17) and (18) using the empirical equation proposed by Khalili and Khabbaz (1998)

$$\chi = \begin{cases} \left(\frac{\psi}{\psi_b}\right)^{-0.55} & \text{when } \psi \geq \psi_b \\ 1 & \text{when } \psi \leq \psi_b \end{cases} \tag{36}$$

where ψ_b = air-entry value (i.e., soil suction beyond which air starts to enter the largest pores in the soil). The ψ_b values are 7 kPa for the PM (Yang et al. 2006), 60 kPa for the DT (Ng et al. 2013), and 70 kPa for the SCL (Gupta et al. 2007). The model parameters used in Eqs. (17) and (18) for the three soils are listed in Table 3 along with the respective R^2 values. Figs. 9–11 show comparisons between the measured and predicted M_R values for the three soils, using Eq. (17) and/or Eq. (18). Fig. 12 shows the overall predictions.

Table 3. Regression Parameters of Eqs. (17) and (18)

Equations	Decomposed tuff	Silty clay loam	Pulverized mudstone
Eq. (17), where M_R, ψ, and σ_d are in kPa			
k_1	274.2	111.5	—
k_2	1.24	1.27	
R^2	0.5	0.26	
Eq. (18), where M_R, ψ, θ_b, and τ_{oct} are in kPa			
k_1	296.8	168.9	744.2
k_2	3.6	5.79	3.12
k_3	−6.69	−9.07	−5.5
R^2	0.96	0.93	0.93

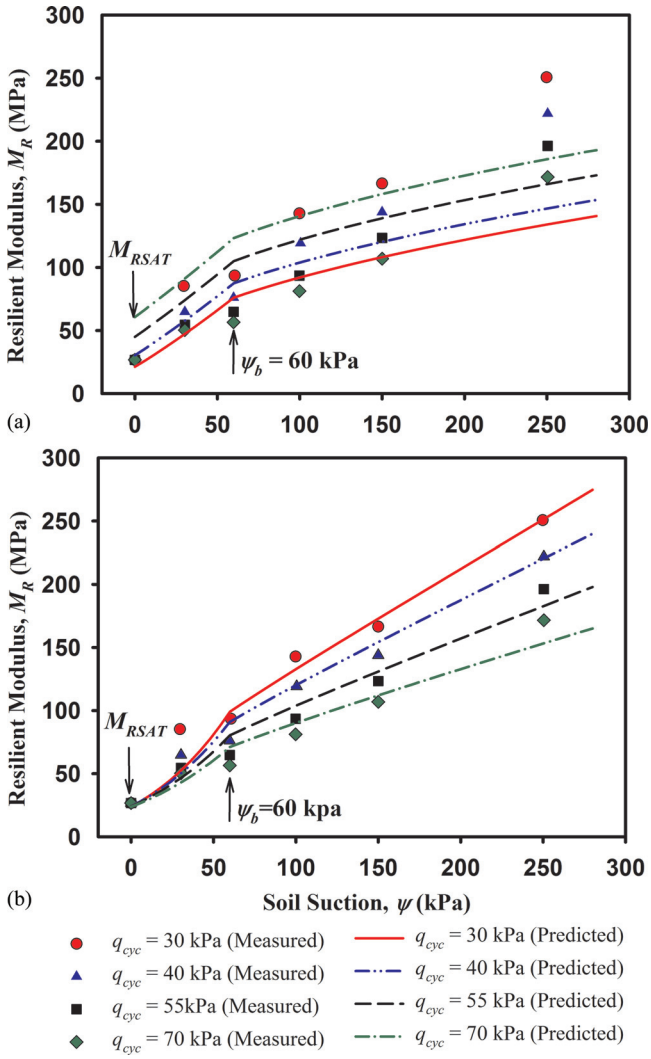


Fig. 9. Comparisons between the variations of the M_R with respect to soil suction of the DT measured by Ng et al. (2013) and predicted by: (a) Eq. (17), $R^2 = 0.5$; (b) Eq. (18), $R^2 = 0.96$

Eqs. (17) and (18) predict nonlinear variations of the M_R with respect to the soil suction for the three soils. It can be observed from Figs. 9–11 that the predicted M_R–ψ curves are steeper when soil suction is lower than the ψ_b, and the predicted curves flatten when

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

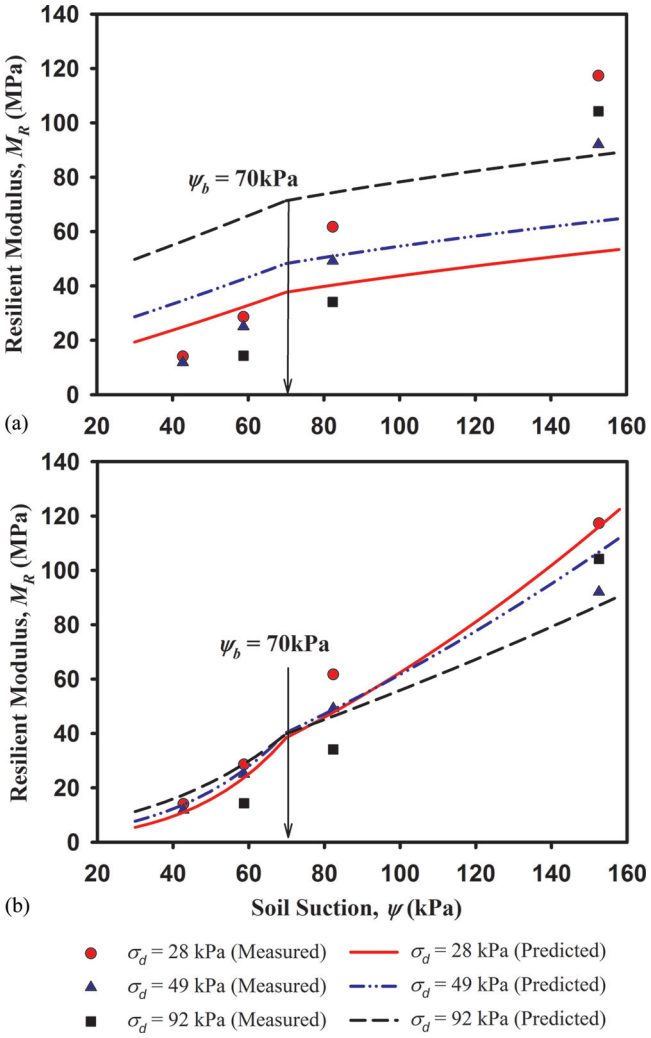


Fig. 10. Comparisons between the variation of the M_R with respect to soil suction of the SCL measured by Gupta et al. (2007) and predicted by: (a) Eq. (17), $R^2 = 0.26$; (b) Eq. (18), $R^2 = 0.93$

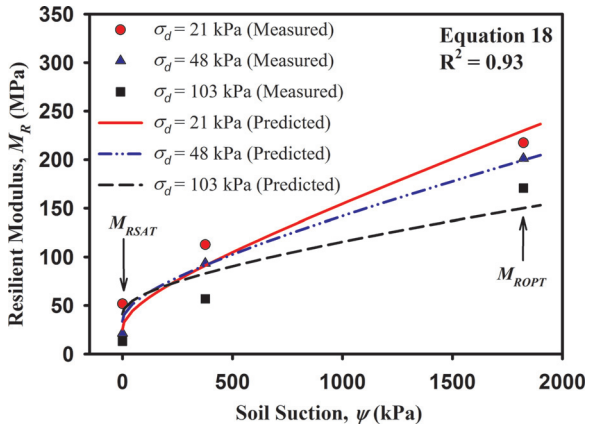


Fig. 11. Comparisons between the variation of the M_R with respect to soil suction of the PM measured by Yang et al. (2005) and predicted by Eq. (18)

soil suction is greater than the ψ_b. Such behavior can be in part attributed to the use of parameter χ, which varies with soil suction, as predicted by Eq. (36).

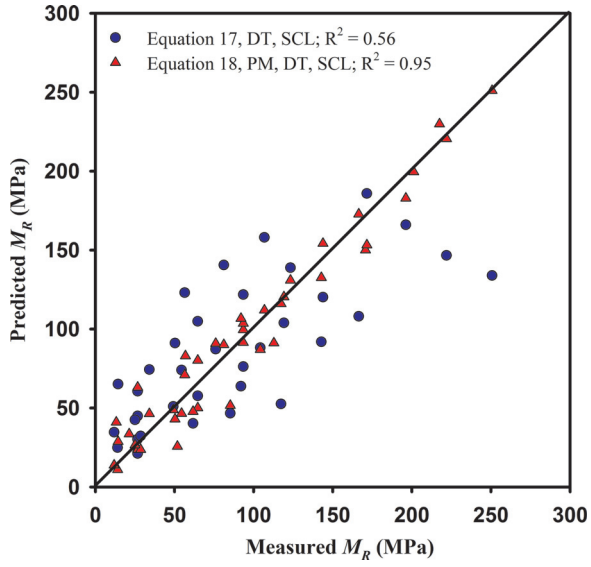


Fig. 12. Overall predictions provided by Eqs. (17) and (18) in comparison with the measured values

The measured M_R values of the three soils increase with increasing soil suction but decrease with increasing applied shearing stresses q_{cyc} and σ_d (Figs. 9–11). Eq. (17) reasonably captures the increase in the M_R with ψ. It predicts increasing M_R with increasing q_{cyc} and σ_d, which is, however, contrary to the measurements. Such results can be attributed to the formulation of Eq. (17), which uses a single model parameter k_2 to predict the behavior of the M_R with respect to both σ_d and ψ. Positive k_2 values derived from regression analysis in this study (Table 3) indicate that M_R increases with both σ_d and ψ. Negative k_2 values, however, will indicate a decreasing trend in the M_R with both σ_d and ψ. The softening behavior of the M_R with increasing applied shearing stresses for fine-grained soils was observed and discussed by some investigators in the literature (Ceratti et al. 2004; Ng et al. 2013; Sivakumar et al. 2013). The hardening behavior of the M_R was also reported (Parreira and Gonçalves 2000; Yang et al. 2008). Eq. (17) is more suitable to predict the M_R–ψ correlations for subgrade soils that exhibit hardening behavior with respect to applied shearing stresses.

Eq. (18) predicts the variations of the M_R with soil suction and shearing stress using model parameters k_2 and k_3, respectively. The positive k_2 values and negative k_3 values derived from the regression analysis (Table 3) reasonably capture the measured behavior of the M_R and, consequently, contribute to better predictions. However, the predicted behavior of the M_R with respect to the σ_d for the PM and SCL are not consistent. The predicted M_R values increase with increasing σ_d at soil suction range lower than ψ_b but decrease with increasing σ_d at soil suction range higher than ψ_b [Figs. 10(b) and 11]. Such behavior may be attributed to the use of bulk stress θ_b ($\theta_b = 3\sigma_c + \sigma_d$) in Eq. (18), which also increases with σ_d.

Comparisons for Constitutive Models from Group C

The variation of the M_R with respect to the ψ is predicted using model parameters α_1 and β_1 in Eq. (27) and model parameter k_3 in Eq. (31). Estimation of the model parameters k_1, k_2, k_3, k_4, α_1, and β_1 in Eq. (27) requires the experimental data of the M_R under various applied confining/shearing stresses and soil suction values. At saturation condition (i.e., $\psi = 0$), Eq. (27) reduces to

Table 4. Regression Parameters of Eqs. (38) and (31)

Equation	Decomposed tuff	Silty clay loam	Pulverized mudstone
Eq. (38), where M_R and ψ are in kPa			
α_1	1352	—	1157
β_1	0.89		0.66
R^2	0.90		0.98
Eq. (31), where M_R, p, q_{cyc}, and ψ are in kPa			
k_1	1.0	1.0	1.0
k_2	–0.65	–0.1	–0.46
k_3	1.01	1.72	0.51
M_0	8.32	0.23	4.07
R^2	0.98	0.92	0.96

$$M_{RSAT} = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(k_4 + \frac{\tau_{oct}}{p_a}\right)^{k_3} \tag{37}$$

Thus, Eq. (27) is simplified as Eq. (38) and used in this study

$$M_R = k_1 p_a \left(\frac{\theta_b}{p_a}\right)^{k_2} \left(k_4 + \frac{\tau_{oct}}{p_a}\right)^{k_3} + \alpha_1 \psi^{\beta_1} = M_{RSAT} + \alpha_1 \psi^{\beta_1} \tag{38}$$

Ng et al. (2013) suggested that the M_R increases linearly with respect to the confining stress (p) for pavement subgrades. Similar suggestion was also proposed by Fredlund et al. (1977). Thus, a value of $k_1 = 1.0$ is used in Eq. (31) assuming the linear M_R–p relationship for the SCL and PM. Other model parameters M_0, k_2, and k_3 in Eq. (31) are derived from the regression analysis. Table 4 summarizes the model parameters used for Eqs. (31) and (38) and their respective R^2 values. Figs. 13–15 show comparisons between the measured M_R values and the predicted M_R values using Eq. (31) and/or Eq. (38) for the PM, DT, and SCL, respectively. Fig. 16 shows the overall predictions.

Both Eqs. (31) and (38) provide excellent predictions ($R^2 \geq 0.9$) according to Witczak et al. (2002) for the experimental data shown in Fig. 16. It can be observed from Figs. 13(a) and 14 that Eq. (38) predicts variations of the M_R regardless of the influence of applied shearing stresses q_{cyc} or σ_d. This in part can be attributed to the second term of Eq. (38) (i.e., $\alpha_1 \psi^{\beta_1}$), which takes into account the influence of the soil suction while neglecting the influence of the applied stresses. Eq. (31), on the other hand, predicts different nonlinear M_R–ψ relationships at various levels of shearing stress for the PM and SCL, as shown in Figs. 13(b) and 15.

Comparisons Presented in Resilient Modulus–Cyclic Stress–Soil Suction Space

Comparisons between the measured M_R of the DT (Ng et al. 2013) and the predicted M_R using the six selected equations are shown in Fig. 17 in the M_R–q_{cyc}–ψ space. The model parameters used in Eqs. (17), (18), (31), and (38) are listed in Tables 3 and 4 [model parameter values used for Eq. (31) were suggested by Ng et al. 2013].

The shape of the surfaces predicted using empirical relationships [i.e., Eqs. (3) and (8)] depends on (1) constant model parameters and (2) referenced M_R values [M_{ROPT} in Eq. (3) and M_{RSAT} in Eq. (8)]. The shape of the surfaces predicted using constitutive models [i.e., Eqs. (17), (18), (27), (31), and (38)] is controlled by (1) methodologies used to account for the contribution of the soil suction,

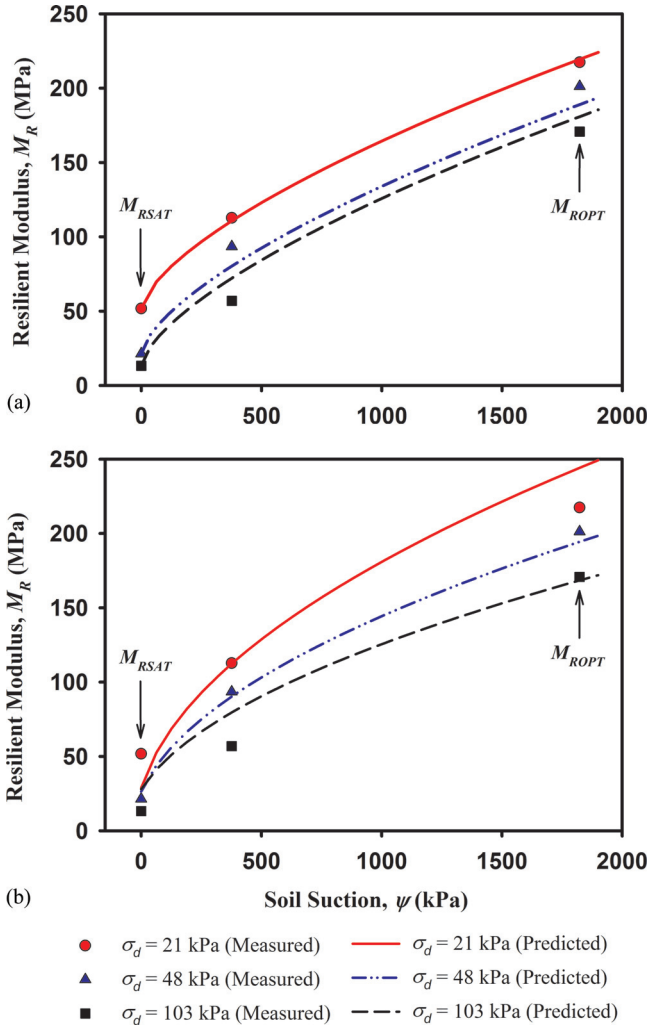


Fig. 13. Comparisons between the variation of the M_R with respect to soil suction of the PM measured by Yang et al. (2005) and predicted by: (a) Eq. (38), $R^2 = 0.98$; (b) Eq. (31), $R^2 = 0.96$

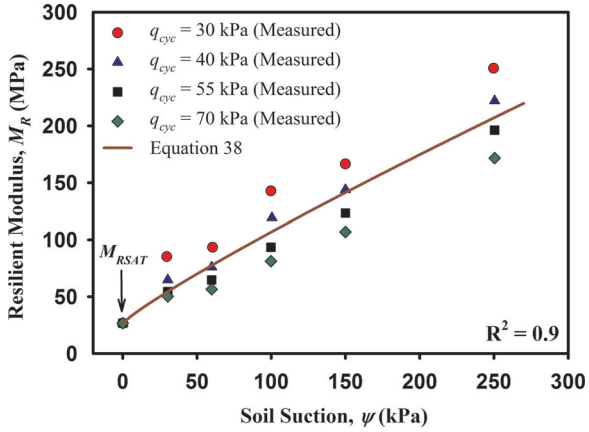


Fig. 14. Comparisons between the variation of the M_R with respect to soil suction of the DT measured by Ng et al. (2013) and predicted by Eq. (38)

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

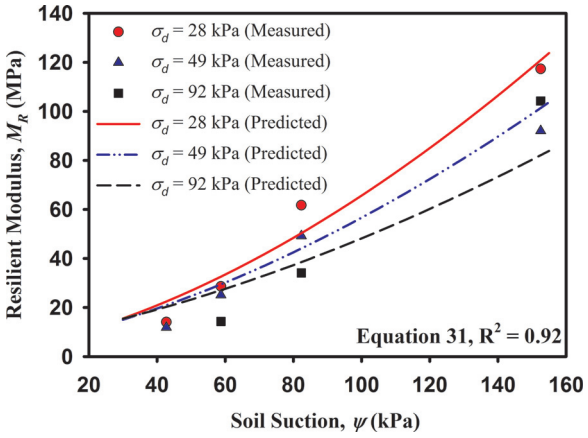


Fig. 15. Comparisons between the variation of the M_R with respect to soil suction of the SCL measured by Gupta et al. (2007) and predicted by Eq. (31)

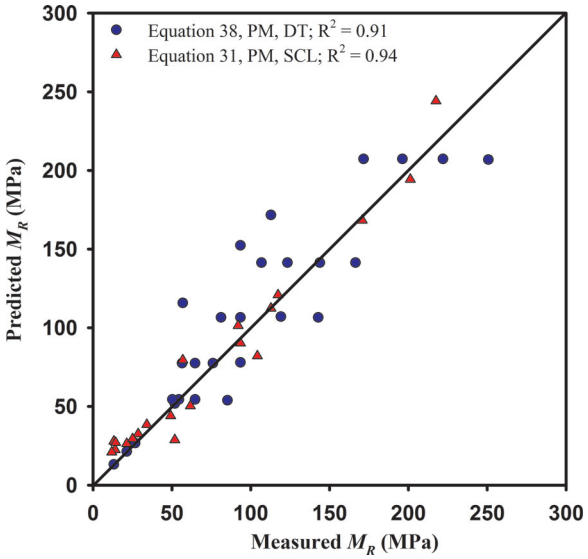


Fig. 16. Overall predictions provided by Eqs. (38) and (31) in comparison with the measured values

(2) applied stresses (θ_b, σ_d, τ_{oct}, p, q_{cyc}, etc.) formulated in the prediction equation, and (3) model parameters (model parameters k_1, k_2, k_3, k_4, M_0, etc.) that need to be calibrated from experimental data.

Generally, constitutive models are flexible in fitting the nonlinear variations of the M_R with respect to the applied stresses and soil suction. Constitutive models extending Bishop's effective stress concept for unsaturated soils predict surfaces with distinctly different shapes prior to and after the air-entry value [Figs. 17(c and d)]. The constitutive models extending the independent stress state variable approach predict smooth surfaces [Figs. 17(e and f)] and better fit the measured M_R–q_{cyc}–ψ relationships for subgrade soils.

It is important to note that reasonable predictions using constitutive models are possible when the model parameters are derived from regression analysis performed on a large experimental database. Different model parameter values are required for the same constitutive model for different soils (Tables 3 and 4). Also, regression parameters may only be reliable for the applied stress and soil suction range of the experimental database. The predicted M_R–ψ correlations using the six selected equations [Eqs. (3), (8), (17), (18), (31), and (38)] show a continuous nonlinear increasing trend with the soil suction (Figs. 9–11, 13–15, and 17). As discussed earlier, the loss of strength and stiffness that occurs at higher soil suction values may not be reasonably predicted.

Summary and Conclusions

Various empirical relationships and constitutive models available in the literature to predict the variation of the M_R with respect to soil suction are summarized in three groups: Group A, empirical relationships; Group B, constitutive models incorporating the soil suction into the applied shearing or confining stresses; and Group C, constitutive models extending the independent stress state variable approach.

Several comments are provided for the summarized equations:

1. Empirical relationships are easy and economical for use in practice, but are only suitable for certain soils for a limited soil suction range. Constitutive models are flexible for considering various factors that influence the M_R and provide reasonably good predictions if the model parameters are calibrated using extensive experimental data. The calibrated model parameters for a particular soil for specific soil suction and applied stress ranges may not be suitable for other soils or even the same soil when it is subjected to different applied stress and soil suction.
2. Most constitutive models proposed to predict the M_R–ψ relationships are revised from conventional constitutive models. However, model parameters for conventional models that were well documented cannot be directly extended into the improved models for prediction.
3. It is important to incorporate the moisture regime (which can be represented by gravimetric/volumetric water contents or degree of saturation) into the constitutive models to rationally predict the M_R–ψ relationships. Neglecting the influence of the moisture content may result in unreasonable predictions of the M_R at a large soil suction range and the inability to account for the influence of hysteresis effects.

Two equations that are widely used from each of the three summarized equation groups are selected to predict the M_R of three subgrade soils with similar soil properties to understand their strengths and limitations. Empirical relationships Eqs. (3) and (8) predict unique M_R–ψ relationships for the three soils regardless of soil type. Eq. (3) tends to overpredict the experimental data, whereas Eq. (8) tends to underpredict the experimental data and is sensitive to the M_{RSAT} values. Constitutive models Eqs. (17) and (18), which incorporate the soil suction into applied stresses using Bishop's effective stress concept for unsaturated soils, predict different M_R–ψ relationships for soil suction ranges before and after the air-entry value. They reasonably predict the nonlinear variation of the M_R with respect to the ψ; but the variation in the M_R with respect to the applied stress is not properly addressed. Constitutive models Eqs. (27) and (31) that extend the independent stress state variable approach are able to reasonably take into account the influence of soil suction and provide reliable predictions within boundary effect and transition zones. However, Eq. (38), which is the simplified form of Eq. (27), does not consider the influence of the applied stress on the M_R–ψ relationships.

Pavement design methodology in recent years has advanced from using empirical or semiempirical methods to computer-aided design using numerical analysis that is based on the mechanistic models. It is important to use suitable and rigorous constitutive models for determining the variations of the M_R with applied stress

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

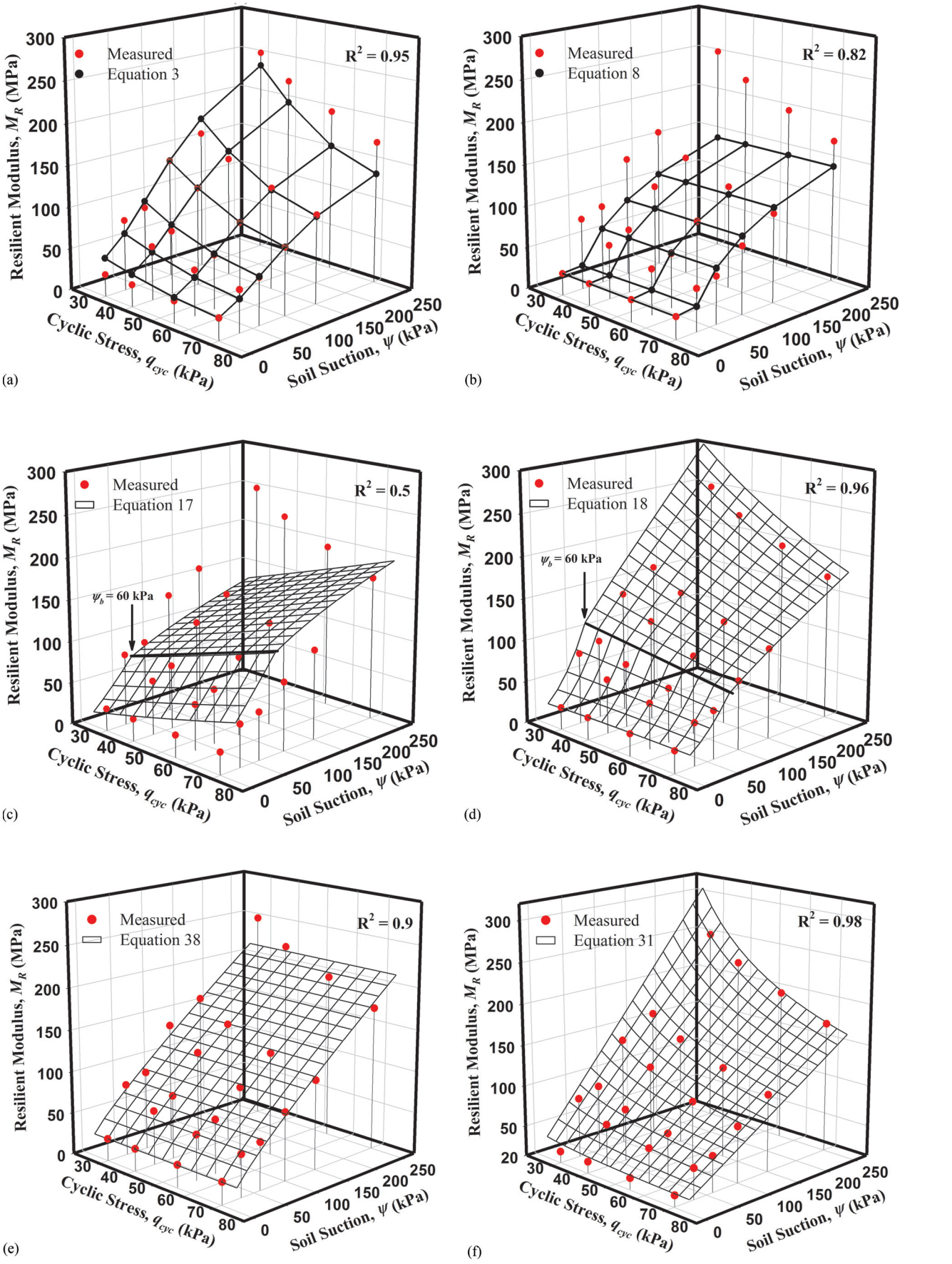


Fig. 17. Comparisons between the M_R of the DT measured by Ng et al. (2013) and predicted by: (a) Eq. (3); (b) Eq. (8); (c) Eq. (17); (d) Eq. (18); (e) Eqs. (27) and (38); (f) Eq. (31) in the M_R–q_{cyc}–ψ space

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

levels and soil suction in the numerical analysis of pavement structure. It is recommended to propose and use constitutive models that provide reasonable predictions based on conventional soil properties (e.g., M_{ROPT}, S_{OPT}) using well-defined model parameters. Such constitutive models can better assist the rational of design of pavement structures.

Acknowledgments

The authors gratefully acknowledge the research funding and financial support from the Natural Sciences and Engineering Research Council (NSERC) of Canada and China Scholarship Council–University of Ottawa Joint Scholarship.

References

AASHTO. (1986). "Standard method of test for resilient modulus of subgrade soils." *T272-82*, Washington, DC.

AASHTO. (1992). "Standard Method of test for resilient modulus of subgrade soils and untreated base/sub-base materials." *T292-91*, Washington, DC.

AASHTO. (2003). "Determining the resilient modulus of soils and aggregate materials." *T307-99*, Washington, DC.

AASHTO. (2008). "Classification of soil and soil-aggregate mixtures for highway construction." *M-145*, Washington, DC.

Abu-Farsakh, M. Y., Mehrotra, A., Mohammad, L., and Gaspard, K. (2015). "Incorporating the effect of moisture variation on resilient modulus for unsaturated fine-grained subgrade soils." *Proc., Transportation Research Board 94th Annual Meeting (No. 15-1149),* Washington, DC.

Alonso, E. E., Gens, A., and Josa, A. (1990). "A constitutive model for partially saturated soils." *Géotechnique*, 40(3), 405–430.

Alonso, E. E., Pinyol, N. M., and Gens, A. (2013). "Compacted soil behaviour: Initial state, structure and constitutive modelling." *Géotechnique*, 63(6), 463–478.

ARA, Inc., ERES Consultants Division. (2004). "Guide for mechanistic–empirical design of new and rehabilitated pavement structures." *Final Rep., NCHRP Project 1-37A*, Transportation Research Board, Washington, DC.

Azam, A. M., Cameron, D. A., and Rahman, M. M. (2013). "Model for prediction of resilient modulus incorporating matric suction for recycled unbound granular materials." *Can. Geotech. J.*, 50(11), 1143–1158.

Ba, M., Nokkaew, K., Fall, M., and Tinjum, J. M. (2013). "Effect of matric suction on resilient modulus of compacted aggregate base courses." *Geotech. Geol. Eng.*, 31(5), 1497–1510.

Bishop, A. W. (1959). "The principle of effective stress." *Tecknish Ukebland*, 106(39), 859–863.

Brown, S. F. (1996). "Soil mechanics in pavement engineering." *Géotechnique*, 46(3), 383–426.

Caicedo, B., Coronado, O., Fleureau, J. M., and Correia, A. G. (2009). "Resilient behaviour of non-standard unbound granular materials." *Road Mater. Pavement Des.*, 10(2), 287–312.

Caliendo, C. (2012). "Local calibration and implementation of the mechanistic-empirical pavement design guide for flexible pavement design." *J. Transp. Eng.*, 10.1061/(ASCE)TE.1943-5436.0000328, 348–360.

Cary, C. E., and Zapata, C. E. (2011). "Resilient modulus for unsaturated unbound materials." *Road Mater. Pavement Des.*, 12(3), 615–638.

Ceratti, J. A., Gehling, W. Y. Y., and Núñez, W. P. (2004). "Seasonal variations of a subgrade soil resilient modulus in southern Brazil." *Transportation Research Record 1874*, Transportation Research Board, Washington, DC, 165–173.

Craciun, O., and Lo, S. C. R. (2010). "Matric suction measurement in stress path cyclic triaxial testing of unbound granular base materials." *Geotech. Testing J.*, 33(1), 33–44.

Doucet, F., and Doré, G. (2004). "Module réversible et coefficient de poisson réversible des matériaux granulaires C-LTPP." *Proc. 57th Canadian Geotechnical Conf. and the 5th Joint CGS/IAH-CNC Conf.,* Québec, Canada.

Drumm, E. C., Reeves, J. S., Madgett, M. R., and Trolinger, W. D. (1997). "Subgrade resilient modulus correction for saturation effects." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)1090-0241(1997)123:7(663), 663–670.

Edil, T. B., and Motan, S. E. (1979). "Soil-water potential and resilient behavior of subgrade soils." *Transportation Research Record 705*, Transportation Research Board, Washington, DC, 54–63.

Edil, T. B., Motan, S. E., and Toha, F. X. (1981). "Mechanical behavior and testing methods of unsaturated soils." *Laboratory shear strength of soil. ASTM STP 740*, American Society for Testing and Materials, West Conshohocken, PA, 114–129.

Edris, E. V., and Lytton, R. L. (1976). "Dynamic properties of subgrade soils, including environmental effects." *Rep. TTI-2-18-74-164-3*, Texas A&M Univ., College Station, TX.

Fredlund, D. G., Bergan, A. T., and Wong, P. K. (1977). "Relation between resilient modulus and stress research conditions for cohesive subgrade soils." *Transportation Research Record 642*, Transportation Research Board, Washington, DC, 73–81.

Fredlund, D. G., Morgenstern, N. R., and Widger, R. A. (1978). "The shear strength of unsaturated soils." *Can. Geotech. J.*, 15(3), 313–321.

Fredlund, D. G., and Xing, A. (1994). "Equations for the soil-water characteristic curve." *Can. Geotech. J.*, 31(4), 521–532.

Fung, Y. C. (1977). *A first course in continuum mechanics*, Prentice Hall, Englewood Cliffs, NJ.

George, K. P. (2004). "Prediction of resilient modulus from soil index properties." *Rep. FHWA/MS-DOT-RD-04-172*, Federal Highway Administration, U.S. Dept. of Transportation, Washington, DC.

Gu, F., Sahin, H., Luo, X., Luo, R., and Lytton, R. L. (2014). "Estimation of resilient modulus of unbound aggregates using performance-related base course properties." *J. Mater. Civ. Eng.*, 10.1061/(ASCE)MT.1943-5533.0001147, 04014188.

Gupta, S. C., Ranaivoson, A., Edil, T. B., Benson, C. H., and Sawangsuriya, A. (2007). *Pavement design using unsaturated soil technology*, Minnesota Dept. of Transportation, St. Paul, MN.

Han, Z., and Vanapalli, S. K. (2015). "Model for predicting the resilient modulus of unsaturated subgrade soil using the soil-water characteristic curve." *Can. Geotech. J.*, 52(10), 1605–1619.

Heath, A. C., Pestana, J. M., Harvey, J. T., and Bejerano, M. O. (2004). "Normalizing behavior of unsaturated granular pavement materials." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)1090-0241(2004)130:9(896), 896–904.

Ho, X. N., Nowamooz, H., Chazallon, C., and Migault, B. (2014a). "Effective stress concept for the effect of hydraulic hysteresis on the resilient behaviour of low traffic pavements." *Int. J. Pavement Eng.*, 16(9), 842–856.

Ho, X. N., Nowamooz, H., Chazallon, C., and Migault, B. (2014b). "Influence of fine content and water content on the resilient behaviour of a natural compacted sand." *Road Mater. Pavement Des.*, 15(3), 606–621.

Hossain, M. S. (2008). "Characterization of subgrade resilient modulus for Virginia soils and its correlation with the results of other soil tests." *Rep. VTRC 09-R4*, Virginia Transportation Research Council, Charlottesville, VA.

Hoyos, L. R., Suescún-Florez, E. A., and Puppala, A. J. (2015). "Stiffness of intermediate unsaturated soil from simultaneous suction-controlled resonant column and bender element testing." *Eng. Geol.*, 188, 10–28.

Jin, M. S., Lee, K. W., and Kovacs, W. D. (1994). "Seasonal variation of resilient modulus of subgrade soils." *J. Transp. Eng.*, 10.1061/(ASCE)0733-947X(1994)120:4(603), 603–616.

Johnson, T. C., Berg, R. L., Chamberlain, E. L., and Cole, D. M. (1986). "Frost action predictive techniques for roads and airfields. A comprehensive survey of research findings." *CRREL Rep. 86–18*, U. S. Army Corps of Engineers, Cold Regions Research and Engineering Laboratory, Hanover, NH.

Khalili, N., Geiser, F., and Blight, G. E. (2004). "Effective stress in unsaturated soils: review with new evidence." *Int. J. Geomech.*, 10.1061/(ASCE)1532-3641(2004)4:2(115), 115–126.

Khalili, N., and Khabbaz, M. H. (1998). "A unique relationship of χ for the determination of the shear strength of unsaturated soils." *Géotechnique*, 48(5), 681–687.

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

Khosravi, A., and McCartney, J. S. (2012). "Impact of hydraulic hysteresis on the small-strain shear modulus of low plasticity soils." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)GT.1943-5606.0000713, 1326–1333.

Khoury, C. N., Khoury, N. N., and Miller, G. A. (2011). "Effect of cyclic suction history (hydraulic hysteresis) on resilient modulus of unsaturated fine-grained soil." *Transportation Research Record 2232*, Transportation Research Board, Washington, DC, 68–75.

Khoury, N., Brooks, R., Khoury, C., and Yada, D. (2012). "Modeling resilient modulus hysteretic behavior with moisture variation." *Int. J. Geomech.*, 10.1061/(ASCE)GM.1943-5622.0000140, 519–527.

Khoury, N. N., Brooks, R., and Khoury, C. N. (2009). "Environmental influences on the engineering behavior of unsaturated undisturbed subgrade soils: Effect of soil suctions on resilient modulus." *Int. J. Geotech. Eng.*, 3(2), 303–311.

Khoury, N. N., and Zaman, M. M. (2004). "Correlation between resilient modulus, moisture variation, and soil suction for subgrade soils." *Transportation Research Record 1874*, Transportation Research Board, Washington, DC, 99–107.

LeKarp, F., Isacsson, U., and Dawson, A. (2000). "State of the art. I: Resilient response of unbound aggregates." *J. Transp. Eng.*, 10.1061/(ASCE)0733-947X(2000)126:1(66), 66–75.

Liang, R. Y., Rabab'ah, S., and Khasawneh, M. (2008). "Predicting moisture-dependent resilient modulus of cohesive soils using soil suction concept." *J. Transp. Eng.*, 10.1061/(ASCE)0733-947X(2008)134:1(34), 34–40.

Loach, S. C. (1987). "Repeated loading of fine grained soils for pavement design." Ph.D. thesis, Univ. of Nottingham, Nottingham, U.K.

Lu, N., and Kaya, M. (2014). "Power law for elastic moduli of unsaturated soil." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)GT.1943-5606.0000990, 46–56.

Lytton, R. L. (1995). "Foundations and pavements on unsaturated soils." *Keynote Address, Proc. 1st Int. Conf. on Unsaturated Soils*, Vol. 3, Balkema, Rotterdam, the Netherlands, 1201–1220.

Moossazadeh, J., and Witczak, M. W. (1981). "Prediction of subgrade moduli for soil that exhibits nonlinear behavior." *Transportation Research Record 810*, Transportation Research Board, Washington, DC, 9–17.

Ng, C. W. W., and Pang, Y. W. (2000). "Influence of stress state on soil-water characteristics and slope stability." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)1090-0241(2000)126:2(157), 157–166.

Ng, C. W. W., and Yung, S. Y. (2008). "Determination of the anisotropic shear stiffness of an unsaturated decomposed soil." *Géotechnique*, 58(1), 23–35.

Ng, C. W. W., and Zhou, C. (2014). "Cyclic behaviour of an unsaturated silt at various suctions and temperatures." *Géotechnique*, 64(9), 709–720.

Ng, C. W. W., Zhou, C., Yuan, Q., and Xu, J. (2013). "Resilient modulus of unsaturated subgrade soil: Experimental and theoretical investigations." *Can. Geotech. J.*, 50(2), 223–232.

Nokkaew, K., Tinjum, J. M., Likos, W. J., and Edil, T. B. (2014). "Effect of matric suction on resilient modulus for compacted recycled base course in postcompaction state." *Transportation Research Record 2433*, Transportation Research Board, Washington, DC, 68–78.

Nowamooz, H., Chazallon, C., Arsenie, M. I., Hornych, P., and Masrouri, F. (2011). "Unsaturated resilient behavior of a natural compacted sand." *Comp. Geotech.*, 38(4), 491–503.

Oh, J. H., Fernando, E. G., Holzschuher, C., and Horhota, D. (2012). "Comparison of resilient modulus values for Florida flexible mechanistic-empirical pavement design." *Int. J. Pavement Eng.*, 13(5), 472–484.

Oh, W. T., and Vanapalli, S. K. (2013). "Interpretation of the bearing capacity of unsaturated fine-grained soil using the modified effective and the modified total stress approaches." *Int. J. Geomech.*, 10.1061/(ASCE)GM.1943-5622.0000263, 769–778.

Oloo, S. Y., and Fredlund, D. G. (1998). "The application of unsaturated soil mechanics theory to the design of pavements." *Proc. 5th Int. Conf. on the Bearing Capacity of Roads and Airfields*, Tapir Academic Press, Trondheim, Norway, 1419–1428.

Parreira, A. B., and Gonçalves, R. F. (2000). "The influence of moisture content and soil suction on the resilient modulus of a lateritic subgrade soil." *GeoEng Int. Conf. on Geotechnical and Geological Engineering*, Melbourne, Australia.

Power, K. C., and Vanapalli, S. K. (2010). "Modified null pressure plate apparatus for measurement of matric suction." *Geotech. Testing J.*, 33(4), 335–341.

Power, K. C., Vanapalli, S. K., and Garga, V. (2008). "A revised contact filter paper method." *Geotech. Testing J.*, 31(6), 461–469.

Puppala, A. J. (2008). *Estimating stiffness of subgrade and unbound materials for pavement design. NCHRP Synthesis 382*, Transportation Research Board, Washington, DC.

Sahin, H., Gu, F., Tong, Y., and Lytton, R. L. (2013). "Unsaturated soil mechanics in the design and performance of pavements." *Keynote Address, Proc. 1st Pan-Am. Conf. on Unsaturated Soils*, CRC Press/Balkema, Rotterdam, the Netherlands.

Salour, F., and Erlingsson, S. (2015). "Resilient modulus modelling of unsaturated subgrade soils: laboratory investigation of silty sand subgrade." *Road Mater. Pavement Des.*, 16(3), 553–568.

Salour, F., Erlingsson, S., and Zapata, C. E. (2014). "Modelling resilient modulus seasonal variation of silty sand subgrade soils with matric suction control." *Can. Geotech. J.*, 51(12), 1413–1422.

Sauer, E. K., and Monismith, C. L. (1968). "Influence of soil suction on behavior of a glacial till subjected to repeated loading." *Hwy. Res. Rec. 215*, Highway Research Board, Washington, DC, 8–23.

Sawangsuriya, A., Edil, T. B., and Benson, C. H. (2009a). "Effect of suction on resilient modulus of compacted fine-grained subgrade soils." *Transportation Research Record 2101*, Transportation Research Board, Washington, DC, 82–87.

Sawangsuriya, A., Edil, T. B., and Bosscher, P. J. (2009b). "Modulus-suction-moisture relationship for compacted soils in postcompaction state." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)GT.1943-5606.0000108, 1390–1403.

Seed, H. B., Chan, C. K., and Lee, C. E. (1962). "Resilience characteristics of subgrade soils and their relation to fatigue failures in asphalt pavements." *Proc., Int. Conf. on the Structural Design of Asphalt Pavements*, ASCE, Reston, VA, 611–636.

Shaqlaih, A., White, L., and Zaman, M. (2013). "Resilient modulus modeling with information theory approach." *Int. J. Geomech.*, 10.1061/(ASCE)GM.1943-5622.0000221, 384–389.

Sivakumar, V., Kodikara, J., O'Hagan, R., Hughes, D., Cairns, P., and McKinley, J. D. (2013). "Effects of confining pressure and water content on performance of unsaturated compacted clay under repeated loading." *Géotechnique*, 63(8), 628–640.

Stolle, D., Guo, P., and Liu, Y. (2009). "Resilient modulus properties of granular highway materials." *Can. J. Civil Eng.*, 36(4), 639–654.

Thom, R., Sivakumar, V., Brown, J., and Hughes, D. (2008). "A simple triaxial system for evaluating the performance of unsaturated soils under repeated loading." *Geotech. Testing J.*, 31(2), 1–8.

Tian, P., Zaman, M. M., and Laguros, J. G. (1998). "Gradation and moisture effects on resilient moduli of aggregate bases." *Transportation Research Record 1619*, Transportation Research Board, Washington, DC, 75–84.

Uzan, J. (1985). "Characterization of granular material." *Transportation Research Record 1022*, Transportation Research Board, Washington, DC, 52–59.

Vanapalli, S. K., Fredlund, D. G., and Pufahl, D. E. (1999). "The influence of soil structure and stress history on the soil-water characteristics of a compacted till." *Géotechnique*, 49(2), 143–159.

Vanapalli, S. K., Fredlund, D. G., Pufahl, D. E., and Clifton, A. W. (1996). "Model for the prediction of shear strength with respect to soil suction." *Can. Geotech. J.*, 33(3), 379–392.

Witczak, M. W. (2003). "Harmonized test method for laboratory determination of resilient modulus for flexible pavement design." *Final Rep. NCHRP 1-28A*, Transportation Research Board, National Research Council, Washington, DC.

Witczak, M. W., Pellinen, T., and El-Basyouny, M. (2002). "Pursuit of the simple performance test for asphalt concrete fracture/cracking." *Asphalt Paving Technol.*, 71, 767–778.

Wu, S., Gray, D. H., and Richart F. E., Jr. (1984). "Capillary effects on dynamic modulus of sands and silts." *J. Geotech. Eng.*, 10.1061/(ASCE)0733-9410(1984)110:9(1188), 1188–1203.

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.

Yang, S. R., Huang, W. H., and Tai, Y. T. (2005). "Variation of resilient modulus with soil suction for compacted subgrade soils." *Transportation Research Record 1913*, Transportation Research Board, Washington, DC, 99–106.

Yang, S. R., Kung, J. H. S., Huang, W. H., and Lin, H. D. (2006). "Resilient modulus of unsaturated cohesive subgrade soils." *Chin. J. Geotech. Eng.*, 28(2), 225–229 (in Chinese).

Yang, S. R., Lin, H. D., Kung, J. H., and Huang, W. H. (2008). "Suction-controlled laboratory test on resilient modulus of unsaturated compacted subgrade soils." *J. Geotech. Geoenviron. Eng.*, 10.1061/(ASCE)1090-0241(2008)134:9(1375), 1375–1384.

Zaman, M., Solanki, P., Ebrahimi, A., and White, L. (2010). "Neural network modeling of resilient modulus using routine subgrade soil properties." *Int. J. Geomech.*, 10.1061/(ASCE)1532-3641(2010)10:1(1), 1–12.

Zapata, C. E., Andrei, D., Witczak, M. W., and Houston, W. N. (2007). "Incorporation of environmental effects in pavement design." *Road Mater. Pavement Des.*, 8(4), 667–693.

Downloaded from ascelibrary.org by University of Sunshine Coast on 10/09/17. Copyright ASCE. For personal use only; all rights reserved.